PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	The impact of transitional care interventions on hospital readmissions in older medical patients: a systematic review
AUTHORS	Fønss Rasmussen, Lisa; Grode, Louise; Lange, Jeppe; Barat, Ishay; Gregersen, Merete

VERSION 1 – REVIEW

REVIEWER	Himali Weerahandi NYU Grossman School of Medicine, USA
REVIEW RETURNED	10-Jul-2020

GENERAL COMMENTS	Overall comments Thank you for the opportunity to review this manuscript. This manuscript is a systematic review on the important topic of the impact of transitional care interventions on readmission rates in older patients. This is an important contribution to the literature on transitional care. I am by no means an expert in systematic reviews, however the methods are extremely detailed and thorough. The authors have demonstrated that they have included the essential items for a systematic review by providing the PRISMA checklist. My only major concern is that the manuscript overall needs copy-editing to improve paragraph organization, sentence structure, and grammar. Minor comments: Introduction: Line 26: I'm unclear what "inadequate patient transition from hospital to home" means---please be more explicit. The rest of the sentence gives specific details on different factors that impact that transition from hospital to home. Results Page 13, line 15: what does patient assessments refer to? Is this an independent evaluation apart from the medical team? Page 13, line 17: does physical exercise refer to regular PT sessions during the inpatient stay, or a PT assessment to determine disposition? Page 13, line 28: "additional health care services"---could you provide additional detail on what this means? Page 13, Synthesis of Results: I'm a bit confused---the first paragraph states that a meta-analysis could not be conducted, but then in lines 45-50 you do present pooled results. I would suggest simply just setting this up better in the first paragraph (i.e "We were not able to conduct a meta-analysis on all 11 studies, but xyz…") Page 18, Intervention intensity: The authors state "Supplementary 7 illustrates a clear difference in the impact between the low and high intensity interventions." But the next sentence states very

	similar RR ranges for both low intensity and high intensity interventions, with both RRs crossing 1. Pages 19-20: I find Table 3 and Figure 2 confusing. These are important, but it's confusing that "strong" and "weak" mean different things for different criteria---I'm assuming "strong" selection bias is "bad" whereas "strong" blinding is good. I think it's fine to use different terms for the different components, maybe you can uses green to signify "good" and red as "bad". Discussion: Page 19: Regarding country of origin, another consideration could be that existing social services are less exhaustive in the United States, and thus the support provided by transitional programs makes a bigger difference there compared to European countries with universal healthcare. The ideas in the "readmission rates prior to the study" may also play a role here if readmission rates are higher in the US than European countries.

REVIEWER	Dr Christina Parker Queensland University of Technology Austraila
REVIEW RETURNED	24-Jul-2020

GENERAL COMMENTS	A well written systematic review - well done. There is just a typo on page 22 - comparison with previous research the heading should be. This will be of interest to a range of health professionals and will be good to see published.

REVIEWER	Peter Goldblatt UCL Institute of Health Equity United Kingdom
REVIEW RETURNED	04-Oct-2020

GENERAL COMMENTS	The authors have gone through a rigorous systematic review process to identify evaluated interventions that met their strict criteria. This has led to few eligible evaluations being identified (11) and even fewer high quality interventions/evaluations (2). In the light of this the conclusions of the paper need to be more strongly qualified by reference to these two limitations. The majority of the methods section of the abstract concerns process. It would be more helpful if it started with and included more information from the section on eligibility criteria, since these are key to what studies were selected for review. Specifically, the authors need to provide a clearer statement of what, in their view, constitutes and eligible TCI in the abstract and what is not eligible. It would be helpful to have some discussion of the appropriateness of the eligibility criteria and some quantification of the number of evaluations excluded by each of criteria i to vii on page 6, lines 22 to 36. The main findings "The most significant impact is seen when interventions: are high intensity, last at least one month and target patient at risk" are not well-justified in the body of the text. The reader is invited to eyeball the forest plots in the supplementary material. What is needed in the text is a summary quantification of the contrast between low and high intensity, different intervention durations and level of targeting. Ideally this would be summary

	estimates of overall effect sizes (e.g. aggregate relative risks) or, failing that, grouping effect sizes and confidence intervals to show that there was a statistical difference. While the overall standard of English is good, there are a few typos. For example, Page 6, line 27 "internal" should be international" Page 23, line 12 "privious" should be "previous"

VERSION 1 – AUTHOR RESPONSE

Reviewer's comments	**Author's response**	**Changes in revised manuscript**
Reviewer 1		
My only major concern is that the manuscript overall needs copy-editing to improve paragraph organization, sentence structure, and grammar.	Thank you very much for your comments. We appreciate your review. Thank you for your suggestion. The manuscript has now been revised by a professional editing and proofreading agency.	
Line 26: I'm unclear what "inadequate patient transition from hospital to home" means--- please be more explicit. The rest of the sentence gives specific details on different factors that impact that transition from hospital to home.	We do agree, that the sentence should be more detailed.	The sentence has been changed to "*Unplanned readmission seems to be related to insufficient discharge planning, and unintended events during discharge and transition such as medication errors and inadequate communication between hospital- and primary care professionals*".
Page 13, line 15: what does patient assessments refer to? Is this an independent evaluation apart from the medical team?	Patient assessment is a broad term. It is now elaborated.	The sentence has been changed to "*patient assessment (Comprehensive Geriatric Assessment, Problem*

		Classification Scheme, Assessment of Patients' Performance of Daily Activities and Assessment of Motor and Process Skills)"
Page 13, line 17: does physical exercise refer to regular PT sessions during the inpatient stay, or a PT assessment to determine disposition?	Patient assessment is a broad term. It will be elaborated.	The sentence has been changed to "*physical exercise (muscle strength, stretch, balance, walking)"*
Page 13, line 28: "additional health care services"---could you provide additional detail on what this means?	Thank you for pointing this out. It will described in details.	The sentence has been changed to *"referral to additional healthcare services (rehabilitation, early specialist follow-up, general practitioner or other community services)"*
Page 13, Synthesis of Results: I'm a bit confused---the first paragraph states that a meta-analysis could not be conducted, but then in lines 45-50 you do present pooled results. I would suggest simply just setting this up better in the first paragraph (i.e "We were not able to conduct a meta-analysis on all 11 studies, but xyz…")	Thank you for the comment. However, we do not agree. The results are not presented as one pooled effect estimate for all studies. The results are merely presented as the numbers of individual study effects according to positive or negative impact on the readmission rate. We modify the sentence to make it more clear.	It has been modified to "*The impact size (RR) from individual studies was calculated using the reported readmission rates from eight studies.*"
Page 18, Intervention intensity: The authors state "Supplementary 7 illustrates a clear difference in the impact between the low and high intensity interventions." But the next sentence states very similar RR ranges for both low intensity and high intensity interventions, with both RRs crossing 1.	The RRs presented are merely ranges of individual study effects. It is not a pooled effect estimate with a RR crossing 1. However, we have changed the sentence to make it more clear.	It has been changed to "*Supplementary 7 illustrates a clear difference. The vast majority of interventions with high intensity reported a positive impact on readmission rates whereas only half of the interventions with a low intensity showed a positive impact. The impacts were larger and statistically significant among high-intensive interventions.*"

Pages 19-20: I find Table 3 and Figure 2 confusing. These are important, but it's confusing that "strong" and "weak" mean different things for different criteria---I'm assuming "strong" selection bias is "bad" whereas "strong" blinding is good. I think it's fine to use different terms for the different components, maybe you can uses green to signify "good" and red as "bad".	The terms are not used differently across the components. "Weak" in all components means bad quality and "strong" means good quality confer supplementary 3. The colures in Table 3 are now changed to make it more comprehensible.	The colures are changed in Table 3
Page 19: Regarding country of origin, another consideration could be that existing social services are less exhaustive in the United States, and thus the support provided by transitional programs makes a bigger difference there compared to European countries with universal healthcare. The ideas in the "readmission rates prior to the study" may also play a role here if readmission rates are higher in the US than European countries.	Thank you for the reflections. They qualify the analysis and will be included. As we do not know the readmission rates prior to the studies in the US or European countries we do not have substance to discuss this.	The sentence has been modified to *"The accessibility, type, character, quality and overall comprehensiveness of healthcare services provided may also play a role.*" No changes has been made to "readmission rates prior to the study"
Reviewer 2		
There is just a typo on page 22 - comparison with previous research the heading should be.	Thank you very much for your comments. We appreciate your review.	The heading is changed to "comparison with *previous* research"
Reviewer 3		
The authors have gone through a rigorous systematic review process to identify evaluated interventions that met their strict criteria. This has led to few eligible evaluations being identified (11) and even fewer	Thank you very much for your comments.	This has been added to the conclusion **"***This, however, should be seen in the light of the fact that only 11 studies met the inclusion criteria and a low*

high quality interventions/evaluations (2). In the light of this the conclusions of the paper need to be more strongly qualified by reference to these two limitations.	This is an important aspect and the conclusion will be modified.	*certainty of evidence according to the GRADE approach*."
The majority of the methods section of the abstract concerns process. It would be more helpful if it started with and included more information from the section on eligibility criteria, since these are key to what studies were selected for review. Specifically, the authors need to provide a clearer statement of what, in their view, constitutes and eligible TCI in the abstract and what is not eligible.	Thank you for pointing this out. The method section in the abstract has been corrected.	The method section is changed to "*Inclusion criteria were: medical patients ≥65 years or mean age in study population of ≥75 years; interventions were transitional care interventions between hospital and home with both pre- and post-discharge components; outcome was hospital readmissions. Studies were excluded if they: included other patient groups than medical patients, included patients with only one diagnosis or patients with only psychiatric disorders. PubMed, The Cochrane Library, Embase, Cinahl, and Web of Science were searched from January 2008 to August 2019. Study selection at title level was undertaken by one author; the remaining selection process, data extraction and methodological quality assessment were undertaken by two authors independently. A narrative synthesis was performed, and effect sizes were estimated."*
It would be helpful to have some discussion of the appropriateness of the eligibility criteria and some quantification of the number of evaluations excluded by each of criteria i to vii on page 6, lines 22 to 36.	Thank you for pointing out the missing aspect of the discussion. The appropriateness of the eligibility criteria is now discussed in the section	This has been added to the section "strength and limitation of this review" "*The eligibility criteria were somewhat narrow, which limited the number of included studies. This, however, was deemed necessary to meet the need for evidence focusing on*

	"strength and limitation of this review". Quantification of excluded studies on full-text level is mentioned in the PRISMA flow chart in Figure 1. Reasons for exclusion covers parts of the criteria i to vii on page 6. Other eligibility criteria were applied when authors screened records by title and abstract. These numbers are unfortunately not available.	*clinical settings where a large number of hospital wards are general medical wards."*
The main findings “The most significant impact is seen when interventions: are high intensity, last at least one month and target patient at risk" are not well-justified in the body of the text. The reader is invited to eyeball the forest plots in the supplementary material. What is needed in the text is a summary quantification of the contrast between low and high intensity, different intervention durations and level of targeting. Ideally this would be summary estimates of overall effect sizes (e.g. aggregate relative risks) or, failing that, grouping effect sizes and confidence intervals to show that there was a statistical difference.	Thank you for the comment. The main findings have been justified in the sub-group analysis "study population" "intervention intensity" "length of support" There are no overall effect estimates as this is a narrative synthesis, and not a meta-analysis. Thus, it is not possible to conduct statistical analysis to show the differences. The forest plots exhibit the difference in effects between the sub-groups.	The text in "intervention intensity has already been changed. The text in "study population" is changed to "*Supplementary 6 exhibits impacts according to the included study population. There is a clear difference in impacts between the groups. In total, 14 of 17 (82%) interventions including "patients at risk" reported positive impacts on readmission rates. In contrast, only five out of nine (56%) interventions including "unselected patients" reported positive impacts. In addition, the positive impacts were larger among interventions including "patient at risk" than among interventions with "unselected patients". Three of the four statistically significant results are found among the "patient at risk" group. However, the 95% Cis are*

		wider among "patients at risk" compared with the "unselected patients" indicating less precise effect estimates". The text in "length of support" is modified to "*Supplementary 8 displays the impact on readmission rates stratified according to the duration of the interventions. A short length of support is associated with less or no impact on readmission rates. A length of support of one month or more is associated with positive, larger and statistically significant impacts.*"
Page 6, line 27 "internal" should be international"	Thank you for comment. The spelling has been corrected.	Studies were excluded if: iii. they included participants with only one medical diagnosis (*International* Classification of Disease -10th edition);
Page 23, line 12 "privious" should be "previous"	Thank you for comment. The spelling has been corrected.	The heading is changed to "comparison with *previous* research"

VERSION 2 – REVIEW

REVIEWER	Peter Goldblatt UCL Institute of Health Equity United Kingdom
REVIEW RETURNED	24-Nov-2020

GENERAL COMMENTS	The authors have addressed the need to clarify the limitations of the studies identified in their systematic review. This paper therefore represents a very honest appraisal of the state of evidence that exists on a very important area of hospital discharge practices. The limited positive findings that it identifies are now appropriately caveated. Where new text has been added it has not always been checked for clarity of the English language used. It would be helpful if this was done, so that the messages around the limitations of the studies identified are clearer to the reader.

VERSION 2 – AUTHOR RESPONSE

Reviewer 3		
Where new text has been added it has not always been checked for clarity of the English language used. It would be helpful if this was done, so that the messages around the limitations of the studies identified are clearer to the reader.	Thank you for the suggestion. After this revision the manuscript has once again been checked for English grammar, spelling and flow	